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COURSE CODE: CSA1716

COURSE NAME: Artificial Intelligence

Artificial Intelligence – Assignment 1

Title: 8-Puzzle Problem as a State-Space Search Problem

1. Problem Statement

The 8-Puzzle Problem consists of a 3×3 board containing eight numbered tiles (1–8) and one empty space. The objective is to rearrange the tiles from an initial configuration to a desired goal configuration by sliding one tile at a time into the empty space. The challenge is to find the shortest sequence of moves that transforms the initial state into the goal state.

2. Objectives

- To understand the concept of state-space representation.
- To represent the 8-puzzle as a search problem.
- To identify the initial state and goal state.
- To apply valid operators for generating successor states.
- To understand how AI search algorithms solve real-world problems efficiently.

3. Initial State

| 2 | 8 | 3 |

| 1 | 6 | 4 |

| 7 | | 5 |

(Blank space represents the empty tile.)

4. Goal State

| 1 | 2 | 3 |

| 8 | | 4 |

| 7 | 6 | 5 |

5. State Representation

A state is represented as a 3×3 **matrix** where each cell contains either:

- A numbered tile (1–8)
- One blank space

Example:

[2 8 3

1 6 4

7 _ 5]

Each arrangement of the board represents a unique state in the search space.

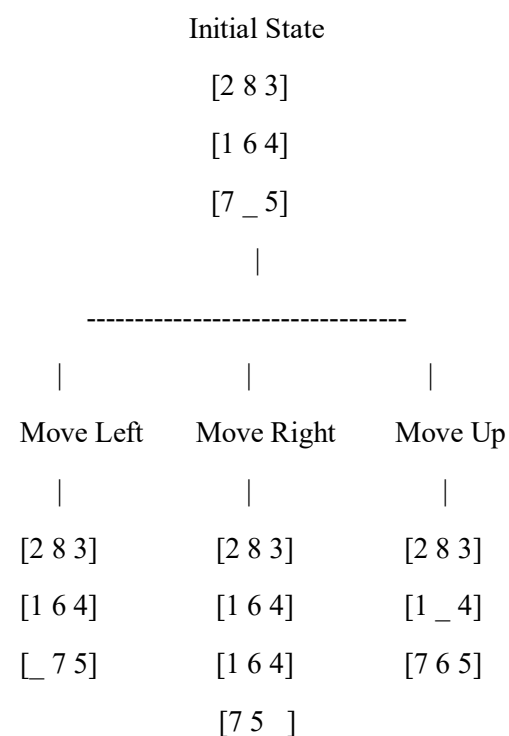
6. Operators / Actions

The blank tile can move in four possible directions, depending on its position:

- **Move Up**
- **Move Down**
- **Move Left**
- **Move Right**

Each move creates a new valid state.

7. State Space Diagram



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Continue Search

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Goal State

8. Applications

1. Robot Motion Planning

- Used to determine the best sequence of movements for robots while avoiding obstacles.

2. Route Navigation Systems

- GPS applications use state-space search to find the shortest or fastest path.

3. Game Artificial Intelligence

- Games like Chess, Sudoku, Rubik's Cube, and Puzzle Solvers use state-space search to determine optimal moves.

4. Automated Planning Systems

- Used in scheduling, logistics, and warehouse automation.

5. Problem Solving in AI

- Demonstrates how algorithms like Breadth-First Search (BFS), Depth-First Search (DFS), A* Search, and Best-First Search explore possible states to reach a solution.

9. References

1. Stuart Russell & Peter Norvig, **Artificial Intelligence: A Modern Approach**, 4th Edition, Pearson.
2. Elaine Rich & Kevin Knight, **Artificial Intelligence**, McGraw-Hill.
3. Nils J. Nilsson, **Artificial Intelligence: A New Synthesis**, Morgan Kaufmann.
4. George F. Luger, **Artificial Intelligence: Structures and Strategies for Complex Problem Solving**, Pearson.